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Three Address Code

Summer 15:

Find the Quadruples and Triples for the following expression
 $a = b * c + b * c$

$$t_1 = b * c$$

$$t_2 = b * c$$

$$a = t_1 + t_2$$

Implementation of Three Address Code

There are 3 representations of three address code:

1. Quadruple
2. Triples

For the following expression: $a = b * c + b * c$

Quadruples

$t_1 = b * c$

$t_2 = b * c$

$a = t_1 + t_2$



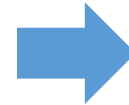
#	op	arg1	arg2	result
0	*	b	c	t_1
1	*	b	c	t_2
2	+	t_1	t_2	a

It is structure with consist of 4 fields namely op, arg1, arg2 and result. op denotes the operator and arg1 and arg2 denotes the two operands and result is used to store the result of the expression. Easy to rearrange code for global optimization. One can quickly access value of temporary variables using symbol table. Contain lot of temporaries. Temporary variable creation increases time and space complexity.

For the following expression: $a = b * c + b * c$

Quadruples

#	op	arg1	arg2	result
0	*	b	c	t_1
1	*	b	c	t_2
2	+	t_1	t_2	a

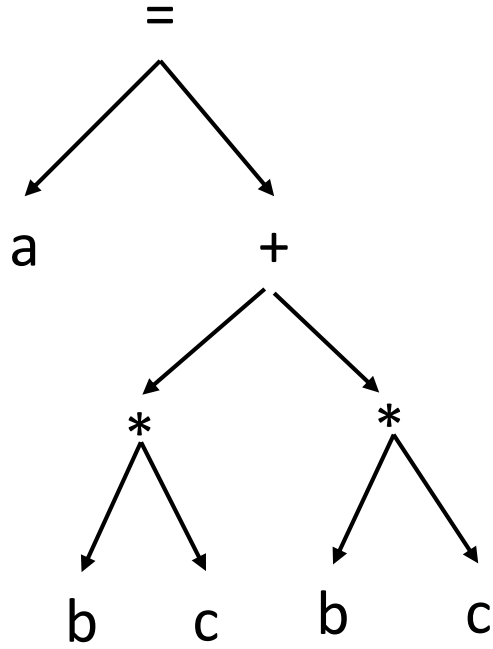


Triples

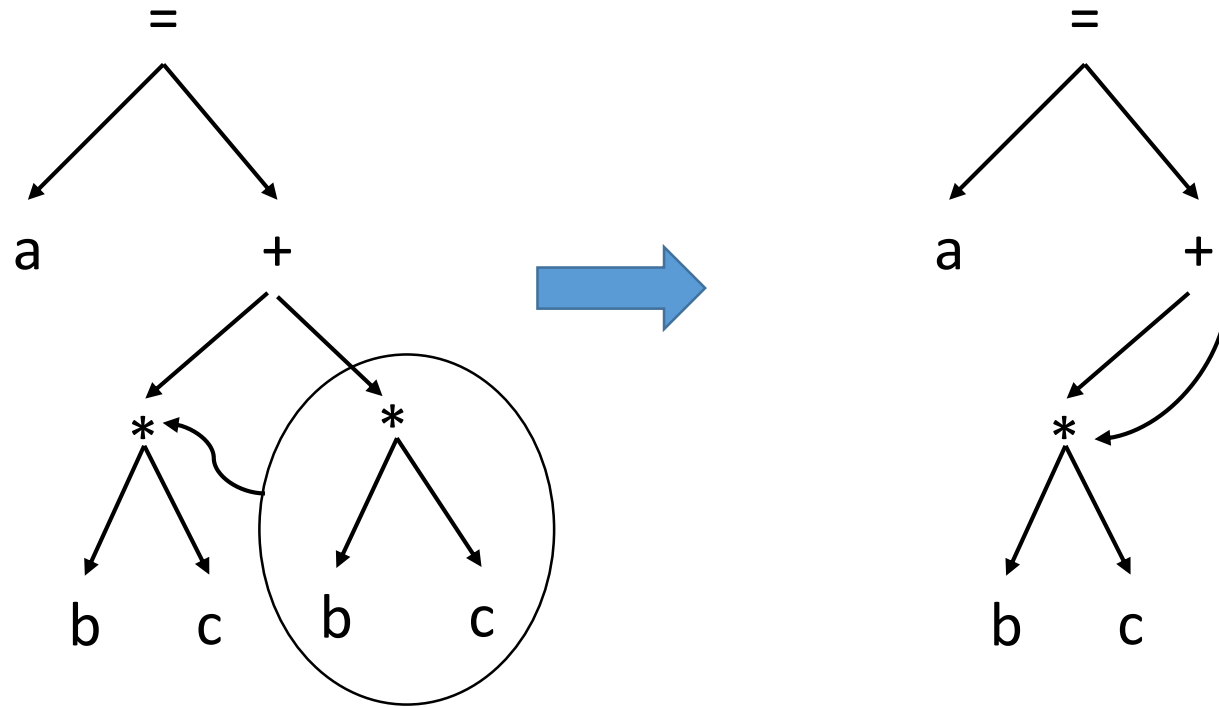
#	op	arg1	arg2
0	*	b	c
1	*	a	(0)
2	+	b	c

Directed Acyclic Graph (DAG)

For the following expression: $a = b * c + b * c$



Syntax directed tree



Directed Acyclic Graph (DAG)

THANK YOU